Adjoint Transport Theory for Rare-Event Sampling in Spin Systems

Technical Manual: Fundamentals, Implementation, and Verified Status

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August 21, 2026

Abstract

Radiation transport solved the problem of sampling rare detector responses decades ago, using the adjoint importance function and the CADIS/FW-CADIS weight-window machinery. The same mathematical object governs rare-event sampling in statistical mechanics, where it is called the *committor*. This manual develops that correspondence from first principles, documents the reference implementation, and states the verified status of every empirical claim. It is written to be self-contained for a reader with no statistical mechanics or transport background — §1 derives the Boltzmann distribution used everywhere below from scratch. Figures are labelled [SCHEMATIC] when they illustrate a concept and [MEASURED] when they plot data measured from the reference code; no schematic curve is presented as a result.

Contents

1 Physics Primer: Spins, Statistics, and Temperature

This section derives everything the rest of the manual assumes: what a spin is, why probability $\propto e^{-E/T}$ at all, what temperature and free energy mean statistically, and why a phase transition exists. A reader already comfortable with the Boltzmann distribution can skip to §2.

1.1 What is a spin?

An electron carries an intrinsic magnetic moment (quantum-mechanical spin) that, once placed in a solid with strong enough coupling to its neighbours, behaves to good approximation as a two-state classical variable: it points "up" or "down" along a fixed axis. This manual works entirely at that classical level — site i carries $s_i \in \{-1, +1\}$, full stop, with no further quantum mechanics anywhere below. Neighbouring spins interact through a coupling J_{ij} contributing energy $-J_{ij}s_is_j$ to the total: for $J_{ij} > 0$ (ferromagnetic) this term is lowest (most negative) when the two spins agree, so the ground state of a uniformly-coupled lattice is *all spins aligned* (two such states, related by flipping every spin at once). The total energy is the Hamiltonian $H(s) = \sum_{\langle i,j\rangle}(-J_{ij}s_is_j)$, summed over neighbouring bonds; §3 makes this precise for the square lattice used throughout.

1.2 Microstates, and why statistics is unavoidable

A *microstate* is one complete assignment of every spin, $s = (s_1, \ldots, s_N)$. Even a modest $L = 16$ lattice has $N = 256$ spins and therefore $2^{256} \approx 10^{77}$ microstates — more than the estimated number of atoms in the observable universe. Nobody solves the exact dynamics of 256 coupled two-state variables and tracks which microstate the system occupies at each instant; statistical mechanics instead asks a different, tractable question: in thermal contact with an environment at temperature T, what is the *probability* of each microstate? This is exactly the same shift in question — from "what happens" to "how likely is it" — that makes rare-event sampling a probability problem rather than a simulation problem (§2).

1.3 The Boltzmann distribution, derived

Put the spin system in contact with a much larger heat reservoir, and treat system+reservoir together as isolated, with fixed total energy $E_{\text{tot}} = E(s) + E_{\text{res}}$. The **fundamental postulate of statistical mechanics** is that an isolated system at fixed total energy occupies every accessible microstate of the *combined* system with equal probability — no microstate is special. The probability that *our* system is found in a particular microstate s is then proportional to the number of ways the reservoir can arrange itself consistent with the leftover energy $E_{\text{tot}} - E(s)$: writing $\Omega_{\text{res}}(E)$ for that count,

$$p(s) \;\propto\; \Omega_{\text{res}}\big(E_{\text{tot}} - E(s)\big). \tag{1}$$

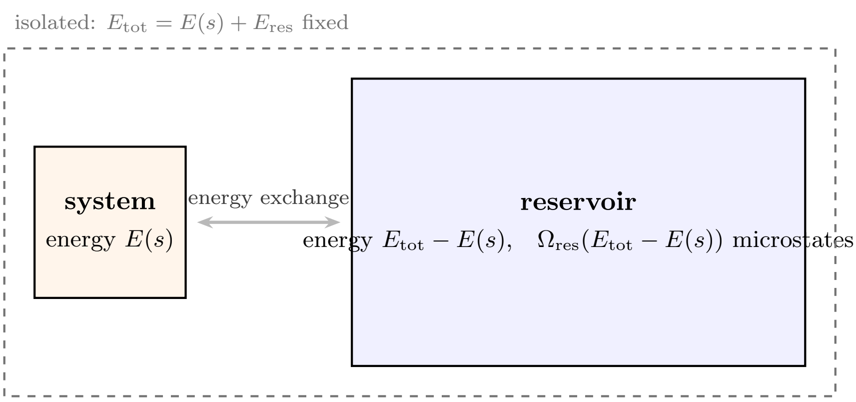

Figure 1: [SCHEMATIC] A small system exchanging energy with a much larger reservoir. Every microstate of the combined (system, reservoir) pair with the same E_{tot} is equally likely; the reservoir's sheer size is what turns that into an exponential law for the system alone.

Because the reservoir is enormous compared to the system, $E(s) \ll E_{\text{tot}}$, so Taylor-expand $\ln \Omega_{\text{res}}$ to first order around E_{tot}:

$$\ln \Omega_{\text{res}}\big(E_{\text{tot}} - E(s)\big) \;\approx\; \ln \Omega_{\text{res}}(E_{\text{tot}}) \;-\; E(s)\, \frac{d \ln \Omega_{\text{res}}}{dE}\bigg|_{E_{\text{tot}}}. \tag{2}$$

Define the reservoir's entropy $S_{\text{res}} = k_B \ln \Omega_{\text{res}}$ (Boltzmann's entropy formula) and *define temperature thermodynamically* by $1/T = dS_{\text{res}}/dE$, evaluated at the reservoir's own energy. Substituting,

$$\ln \Omega_{\text{res}}\big(E_{\text{tot}} - E(s)\big) \;\approx\; \text{const} - \frac{E(s)}{k_B T}, \tag{3}$$

and exponentiating gives the **Boltzmann distribution**

$$p(s) \;\propto\; e^{-E(s)/k_B T}. \tag{4}$$

This is derived, not assumed: it follows from nothing more than "every accessible microstate of an isolated system is equally likely" plus "the reservoir is huge." This manual uses the standard lattice-model convention $k_B = 1$ throughout, so temperature is measured directly in units of energy and Eq. (4) is exactly the $p(s) \propto e^{-H(s)/T}$ asserted at the start of §2.

1.4 The partition function and free energy

Normalising Eq. (4) requires summing over all 2^N microstates,

$$Z = \sum_s e^{-E(s)/T}, \qquad p(s) = \frac{e^{-E(s)/T}}{Z}. \tag{5}$$

Z, the **partition function**, is the normalising constant that "partitions" total probability across every microstate — and, not coincidentally, is exactly the same brute-force sum over 2^N terms that Eq. (9)'s cost argument shows becomes intractable for anything but the smallest lattices. The **free energy**

$$F = -T \ln Z \tag{6}$$

packages everything thermodynamics needs from Z into one scalar, and satisfies the identity $F = \langle E \rangle - TS$: the system's equilibrium state is the one balancing two competing tendencies, minimising energy $\langle E \rangle$ (which favours *order*) against maximising entropy S (which favours *disorder*), with T setting the exchange rate between them. This is the same $F(\lambda)$ plotted schematically against the committor in Fig. 9 — a free-energy *landscape* is just F evaluated along a coordinate $\lambda(s)$ rather than summed over all of it.

1.5 Order, disorder, and why a phase transition exists

At low T, the TS term in $F = \langle E \rangle - TS$ is negligible and the system minimises energy alone: spins align, giving the *ordered* (ferromagnetic) phase. At high T, TS dominates and the system maximises entropy instead: spins are effectively random, $m \approx 0$, the *disordered* (paramagnetic) phase. In the $N \to \infty$ limit these two regimes are separated by a sharp phase transition at T_c (finite lattices smooth this into a crossover); for the 2D ferromagnet on a square lattice this happens exactly at $T_c \approx 2.269$ (§3.2, [10]).

Close to T_c, the competition between order and disorder produces correlated domains at every length scale simultaneously, and single-spin-flip dynamics (Metropolis or Glauber, updating one spin at a time) can only relax these domains locally, one flip at a time — so the time needed to decorrelate, τ_{corr}, *diverges* as $T \to T_c$ (**critical slowing down**). This is measured directly, not asserted: integrated autocorrelation time spikes to $\tau_{\text{int}} = 46.1$ sweeps at T_c versus 1.5–3 sweeps away from criticality (§7.2). This is exactly why every result in this manual is run at $T = 2.6$, comfortably above T_c: it keeps "how long the chain takes to mix" ($\tau_{\text{corr}} \approx 1$ sweep, §6.1) cleanly separated from "how rare is the target event" (set by k, §2.3) — two different kinds of difficulty this manual is careful never to let one masquerade as the other.

2 Fundamentals: The Problem

2.1 A rare event, concretely

Place N magnetic moments ("spins") on a square lattice, each pointing up or down: $s_i \in \{-1, +1\}$. The system has energy $H(s)$, and at temperature T it visits configuration s with probability $\propto e^{-H(s)/T}$ — the Boltzmann distribution derived in §1.3. Left alone it spends essentially all its time in *typical* configurations — call that set A, the *bulk*: everything an equilibrium run visits without effort.

Definition 1 (Rare event)**.** *Let $(s_t)_{t\geq 0}$ be a Markov chain on state space Ω with stationary distribution $\pi_s(s) \propto e^{-H(s)/T}$ (here: Glauber/Metropolis spin-flip dynamics). A* rare event *is a subset $B \subset \Omega$, disjoint from the bulk A, with stationary probability $\pi := \pi_s(B)$ small enough that i.i.d. sampling from π_s needs $M \gtrsim 1/\pi$ draws (Eq. (9)) to observe B even once at useful precision. Rarity is therefore not an intrinsic property of B — it is a relation between π and the sampling budget M actually available. A set with $\pi = 10^{-3}$ is rare at $M = 10^4$ and routine at $M = 10^8$.*

Concretely in this manual, B is a lower tail of the energy distribution:

$$B = \{s : E(s)/N \leq h\}, \tag{7}$$

with h well below anything ordinary sampling produces. We want $\pi = \mathbb{P}(s \in B)$, and π may be 10^{-5} or smaller.

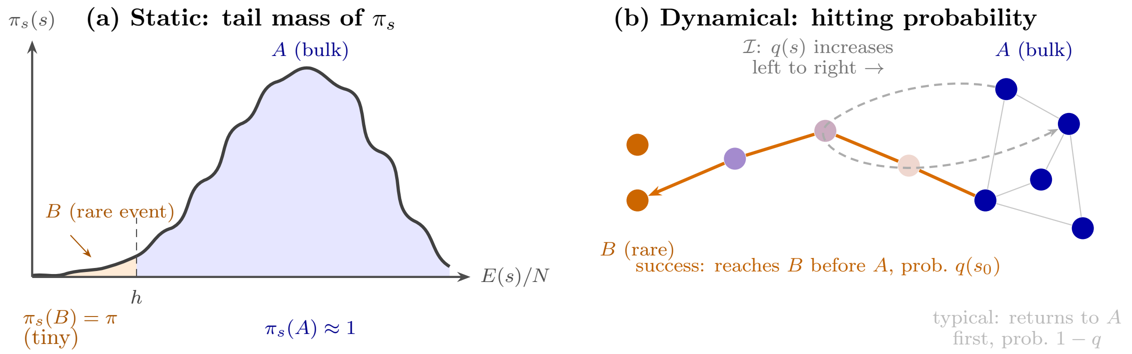

Figure 2: [SCHEMATIC] Two readings of the same rare event B (Def. 1), matching the static/dynamical distinction of §2.3 below. (a) π as tail mass of the stationary distribution — answerable by enumeration alone, no dynamics. (b) $\hat{\pi}$ as this manual actually reports it: a hitting probability, the committor $q(s) = \mathbb{P}(\tau_B < \tau_A \mid s_0 = s)$ of §4 (Eq. (12)). Node shade in (b) encodes q, from bulk ($q \approx 0$) to target ($q = 1$); most trajectories that leave A relax back into it (dashed), which is exactly why brute force is expensive (Fig. 3) and why a good importance function is worth learning (§7).

2.2 Two notions of "rare": static tail versus dynamical hitting probability

Definition 1 is deliberately silent on *how* π is measured, and this manual uses two different answers that must not be confused.

Static: $\pi_s(B)$ is a property of the equilibrium distribution alone — a timeless question about how much Boltzmann weight sits in B, answerable in principle by direct enumeration with no dynamics at all. This is exactly what the $L = 4$ exact-enumeration gate (§5.2) computes, and it is what Eq. (9)'s brute-force argument assumes it is estimating.

Dynamical: every $\hat{\pi}$ reported in a results table elsewhere in this manual (§8) is *not* that. It is a *hitting probability*: starting a trajectory in the bulk A, what is the probability it reaches B before returning to A? This is the committor $q(s) = \mathbb{P}(\tau_B < \tau_A \mid s_0 = s)$ formalized in §4, and Weighted Ensemble (§5) is built to estimate exactly this quantity efficiently, not the static tail directly.

The two coincide under a standard metastability assumption: if dynamics mixes quickly *within* A relative to the rate of leaving it toward B, then the long-run fraction of time spent in B (the static

$\pi_s(B)$) and the per-attempt hitting probability from a freshly-drawn bulk state (the dynamical q-based rate) become proportional, and both are routinely called "π" without further comment — the convention this manual follows too. This holds throughout the $T = 2.6 > T_c$ regime used everywhere below, where the bulk is a single connected basin. It is not guaranteed to hold at low temperature, where A itself could fragment into multiple metastable sub-basins separated by their own barriers — a regime this manual does not probe (§12).

2.3 The rarity ladder used throughout this manual

Every empirical result below expresses rarity depth as an integer k (the `-beyond` flag), not a raw energy threshold, because a raw threshold is not portable across lattice size, temperature, or disorder realization. Given a pilot run's population, let $E_{\text{min}}^{\text{pilot}}$ be the deepest energy per spin reached and μ the bulk mean energy per spin over that same population. The target threshold is

$$h_k = E_{\text{min}}^{\text{pilot}} - k\Delta, \qquad \Delta = \frac{\mu - E_{\text{min}}^{\text{pilot}}}{10}, \qquad B_k = \{s : E(s)/N \le h_k\}, \tag{8}$$

i.e. k steps beyond the pilot's own most extreme observation, in units of one tenth of the pilot's observed dynamic range. Because h_k is decreasing in k, the family is nested, $B_1 \supset B_2 \supset B_3 \supset \cdots$ — exactly the structure a multi-threshold method like FW-CADIS (§7.6) is built to exploit, and the reason milestoning (§8.1) can define "progress" as position within this ladder rather than a single fixed target.

k is a genuine rarity dial: §6 measures, at fixed $L = 16$, $T = 2.6$, roughly one order of magnitude drop in $\hat{\pi}$ per step of k ($k = 1$: $\hat{\pi} \approx 2 \times 10^{-5}$; $k = 2$: $\hat{\pi} \approx 10^{-6}$; $k = 3$: statistics too starved to estimate reliably at matched cost, Table 4) — and it is also where the ladder-difficulty discipline of §6 bites: too large a k places B_k outside what any method can resolve at the compute available, and every comparison degrades to noise-versus-noise.

2.4 Why brute force fails, quantitatively

Draw M independent configurations and count hits. The count is Poisson with mean $M\pi$, so

$$\frac{\sigma_{\hat{\pi}}}{\pi} \approx \frac{1}{\sqrt{M\pi}}. \tag{9}$$

Ten percent error at $\pi = 10^{-6}$ needs $M \approx 10^8$ samples. At $L = 16$ each sample costs one sweep $= N = 256$ spin updates, so that is 2.6×10^{10} spin updates for one threshold.

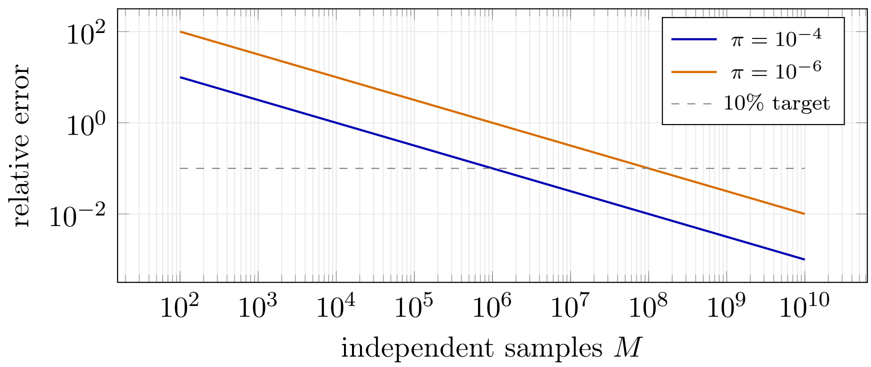

Figure 3: [SCHEMATIC] Exact Poisson relation (9), not fitted data. Each decade of rarity costs a decade of samples. This is the wall the project exists to get around.

2.5 The idea behind every solution

All enhanced-sampling methods make the same move: **spend effort where it matters**. If we knew, for each configuration s, the probability that a trajectory from s reaches B before returning to the bulk A, we could steer computation toward promising configurations. That probability is the committor $q(s)$. The rest of this manual is about why q is the optimal guide, why transport theory already has it under another name, and how to approximate it.

3 The Lattice and the Model

3.1 Geometry and Hamiltonian

On an $L \times L$ lattice with periodic boundaries, $N = L^2$,

$$H(s) = -\sum_{\langle i,j \rangle} J_{ij} s_i s_j, \qquad E(s)/N = H(s)/N, \tag{10}$$

with $\langle i, j \rangle$ over nearest-neighbour pairs and periodic wrapping $s_{(x+L) \bmod L,\, y} = s_{x,y}$, $s_{x,\,(y+L) \bmod L} = s_{x,y}$.

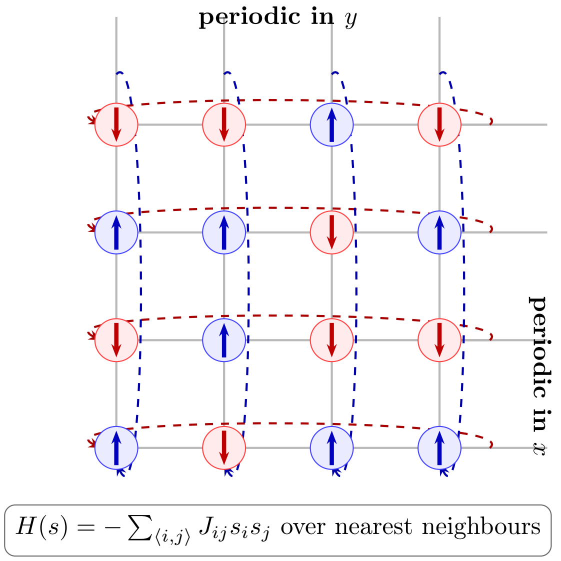

Figure 4: [SCHEMATIC] The $L \times L$ lattice, drawn at $L = 4$. Blue = spin up, red = spin down, solid lines are coupling bonds J_{ij}, dashed arrows are the periodic identifications.

3.2 Two coupling choices

- **Ferromagnet**: $J_{ij} = +1$ uniformly. Ordering transition at $\beta_c = \frac{1}{2} \ln(1 + \sqrt{2}) \approx 0.4407$, i.e. $T_c \approx 2.269$. Magnetisation m is an obviously informative reaction coordinate.
- **Edwards–Anderson spin glass**: J_{ij} drawn independently and **bimodally**, $J_{ij} \in \{-1, +1\}$ with equal probability, quenched by a fixed seed. Low-energy states are disordered, $m \approx 0$ regardless of energy, and a hand-picked coordinate is expected to fail.

All results use $T = 2.6$ (above T_c), $L = 16$ for full scale, $L = 4$ for exact-enumeration gates.

Correction to earlier drafts

An earlier version of this manual stated $J_{ij} \sim \mathcal{N}(0, \sigma_J^2)$. The reference implementation uses `rng.choice([-1.0, 1.0])` — the $\pm J$ bimodal model. The two differ in ground-state degeneracy and universality class, so the distinction is not cosmetic.

3.3 Frustration: why this landscape is hard

A plaquette is *frustrated* when the product of its four bonds is negative: no spin assignment satisfies all four simultaneously.

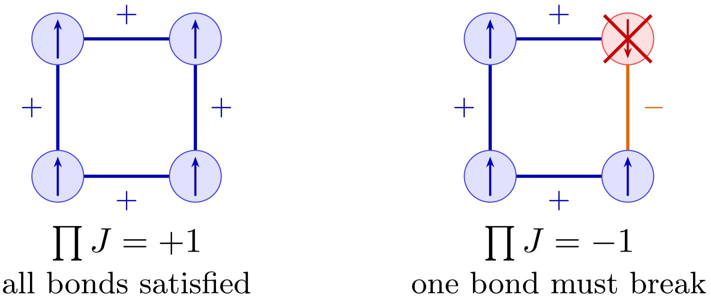

Figure 5: [SCHEMATIC] Left: all four bonds ferromagnetic, product $+1$, a satisfying assignment exists. Right: one antiferromagnetic bond makes the product -1; whichever spins are chosen, at least one bond is unsatisfied. Frustration is what makes the energy landscape rugged.

For the disorder sample used throughout ($L = 16$, seed 0), direct measurement gives:

antiferromagnetic bonds ($J = -1$)	233 of 512
frustrated plaquettes	**126 of 256 (49.2%)**

Nearly half the lattice cannot be locally satisfied. There is no growing domain and no visible order parameter — which is why magnetisation carries little information here, and why a *learned* coordinate is worth the effort.

3.4 From two dimensions to three: what would change

Everything above is stated for the 2D square lattice this manual actually runs. None of it was tried in 3D. This subsection makes precise what would have to change and, more usefully, what would not — because the two lists are not the same size.

Geometry. A simple-cubic lattice replaces the square one: $N = L^3$ sites, each with 6 nearest neighbours (coordination number $z = 6$) instead of 4, periodic wrapping in all three directions. The Hamiltonian keeps the identical form, $H(s) = -\sum_{\langle i,j \rangle} J_{ij} s_i s_j$, summed over the now-larger bond set. Nothing about the physics setup of §3–§3.2 needs rethinking, only re-indexing.

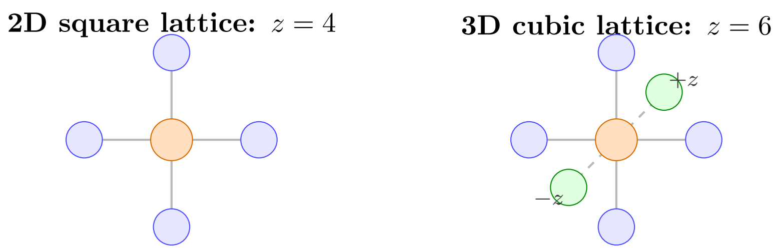

Figure 6: [SCHEMATIC] Nearest-neighbour coordination: a 2D square lattice (left) gives each spin $z = 4$ neighbours (blue); a 3D cubic lattice (right) adds a third axis (green, dashed bonds shown foreshortened for depth), giving $z = 6$. The Hamiltonian's form, $H(s) = -\sum_{\langle i,j \rangle} J_{ij} s_i s_j$, is unchanged — only the bond count per site (orange, centre) grows.

The exact reference disappears. The 2D square-lattice ferromagnet has Onsager's exact solution, $T_c \approx 2.269$ [10] — the number this manual's own $T = 2.6$ choice is pinned against (§1.5). No exact solution exists for the 3D cubic Ising model; T_c is known only from high-precision numerics, currently $T_c/J \approx 4.5115$ (from $K_c = J/k_B T_c = 0.2216544(3)$) via large-scale cluster Monte Carlo

[11]. Any 3D extension of this manual would inherit a numerical, not exact, temperature scale to calibrate against.

The $L = 4$ exact-enumeration gate breaks, and this is the serious obstacle. Every validation claim in this manual (§5.2 and throughout) rests on checking $L = 4$ against direct enumeration of all 2^N microstates. In 2D, $L = 4$ means $N = 16$ and $2^{16} = 65536$ — trivial. In 3D, $L = 4$ means $N = 64$ and $2^{64} \approx 1.8 \times 10^{19}$ — utterly infeasible. Keeping exact enumeration tractable in 3D means dropping to roughly $L = 2$ ($N = 8$, $2^8 = 256$), a lattice almost certainly too small to exercise the WE/committor machinery the way $L = 4$ does in 2D (too few bins, too little structure). This is not a matter of more compute; it is the validation *strategy itself* that does not survive the dimension change, and any 3D extension would need a different anchor (series expansions, or a published high-L numerical benchmark, in place of exact enumeration).

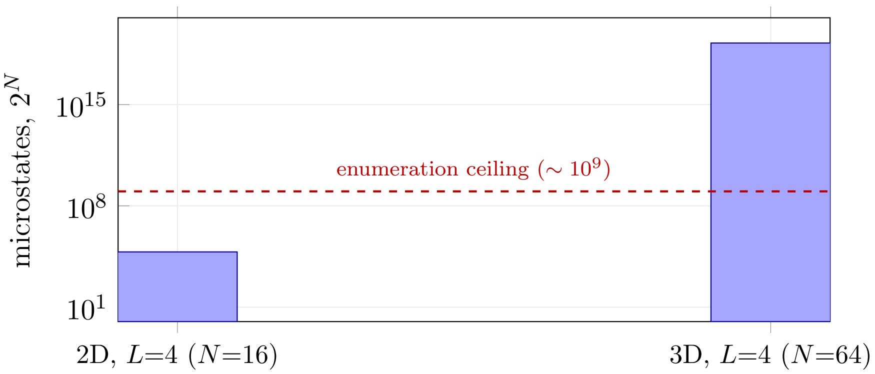

Figure 7: [SCHEMATIC] Exact-enumeration cost at $L = 4$: trivial in 2D ($2^{16} = 65536$ states), utterly infeasible in 3D ($2^{64} \approx 1.8 \times 10^{19}$ states) — fourteen orders of magnitude beyond any practical enumeration budget. This is why the $L = 4$ validation strategy used throughout this manual (§5.2) cannot simply be re-run one dimension higher.

The spin-glass physics changes qualitatively, not just quantitatively. The 2D Edwards–Anderson spin glass used throughout this manual has *no finite-temperature ordering transition* — it orders, if at all, only as $T \to 0$ [12]. Every run in this manual, however cold, is therefore formally still in a single (extremely rough) phase: frustration produces a rugged landscape without a genuine thermodynamic phase boundary to cross. The 3D $\pm J$ Edwards–Anderson model is different in kind: it has a genuine finite-temperature spin-glass transition, $T_g/J \approx 1.10$ (from $\beta_c = 0.908(2)$) [13]. A 3D extension could therefore ask the rare-event question on either side of a real phase boundary — a qualitatively different and arguably richer problem than anything tested here, not merely a bigger version of it.

Sampling cost scales faster with L. One Monte Carlo sweep costs N spin updates (§2), and $N = L^3$ grows faster than the 2D $N = L^2$ used everywhere in this manual's cost estimates. Matching the $N = 256$ scale used throughout (2D, $L = 16$) needs only $L \approx 6$ in 3D ($L = 6 \Rightarrow N = 216$); running 3D at the same $L = 16$ used everywhere here instead means $N = 4096$, a $16\times$ larger per-sweep cost, and correspondingly larger WE populations at fixed bin occupancy. This changes the achievable-scale calculus quantitatively even before asking whether the harder physics helps or hurts variance reduction.

What would *not* need to change. This is the encouraging half. The rare-event sampling machinery this manual is actually about — the committor identity (§4), Weighted Ensemble split/merge (§5), milestoning labels (§7.3), and FW-CADIS allocation (§7.6) — operates entirely on the energy/-committor coordinate and the Markov transition kernel, neither of which knows or cares whether the underlying lattice is 2D or 3D. Only lattice generation, neighbour-list construction, and the VAN's autoregressive raster order (§7.2) would need to change; none of §5–§7 would need modification in principle.

Concretely: how the rare event would be sought. The recipe is identical to §2–§7, re-run on the cubic lattice. A pilot run at the chosen T (comfortably above the relevant T_c or T_g, §1.5) gives $E_{\min}^{\text{pilot}}$ and μ; the rarity ladder B_k of Eq. (8) is defined from those two numbers exactly as before — the formula is written entirely in terms of pilot-run statistics, so it *re-calibrates itself* to whatever energy landscape the cubic lattice produces, with no change to the formula, only to which lattice generates the pilot. Weighted Ensemble bins along the same coordinate choices (energy E, or a trained $I_\theta(s)$) using the identical split/merge rule (§5); milestoning labels (§7.3) place intermediate targets along the same ladder. The one piece that does change shape is I_θ's input: a network trained on a 2D $L \times L$ raster must instead consume an $L \times L \times L$ voxel grid (or its flattened $N = L^3$ vector) — a change to the network's input geometry, not to what it is trained to predict or how WE uses its output.

Concretely: removing critical slowing down in 3D. The VAN strategy of §7.2 sidesteps critical slowing down entirely because it never runs local-update dynamics: it draws whole configurations one variable at a time from a learned autoregressive factorisation $p_\theta(s) = \prod_i q_\theta(s_i \mid s_{<i})$, so every sample is independent by construction regardless of how close T is to T_c. That argument never references dimensionality, so it carries over to 3D unchanged in substance: the raster scan over the $L \times L$ grid (Fig. 13) becomes a scan over the $L \times L \times L$ volume — concretely, sweep through L two-dimensional layers in sequence, raster-scanning each layer before moving to the next, so "earlier in the scan" still means "earlier" in one consistent global order — and the 2D masked convolutions (PixelCNN-style) generalise directly to 3D masked ("causal") convolutions, an established technique for autoregressive models over volumetric data. Nothing about the variational training objective changes. Critical slowing down itself is not a 2D peculiarity: local-update dynamics relax collective modes near *any* continuous transition, 2D or 3D, and recent five-loop field-theoretic estimates of the dynamic critical exponent for Model A (single-spin-flip) dynamics give $z \approx 2.14(2)$ in 2D against $z \approx 2.0235(8)$ in 3D [14] — both close to the diffusive value $z \approx 2$, so the VAN's advantage over local updates is not expected to be dramatically larger or smaller in 3D than the $\tau_{\text{int}} = 46.1$-sweep spike already measured at T_c in 2D (§7.2); it removes the same qualitative bottleneck for the same reason.

This project did not run 3D (§12 lists it as an untouched push-to-scale direction); this subsection exists to state precisely where the real obstacles are — the broken exact-enumeration gate above all — not to claim the extension has been tried.

4 The Transport–Statistical Mechanics Correspondence

4.1 Two fields, one function

In radiation shielding one asks: given a source and a detector behind a thick shield, what fraction of particles reach the detector? Almost none do — the same rare-event structure. Transport theory answers with the *adjoint flux* $\psi^\dagger$, defined by $H^\dagger\psi^\dagger = \sigma_d$ with a source placed *at the detector*. Its physical reading: $\psi^\dagger(P)$ is the expected contribution to the tally of one particle introduced at phase-space point P — its *importance*.

In spin dynamics the committor is

$$q(s) = \mathbb{P}(\tau_B < \tau_A \mid s_0 = s), \qquad \tau_X = \min\{t \geq 0 : s_t \in X\}, \tag{11}$$

satisfying the discrete backward Kolmogorov equation on intermediate states $\mathcal{I} = \Omega \setminus (A \cup B)$,

$$\sum_{s'} K(s \to s')\, q(s') = q(s) \quad \forall s \in \mathcal{I}, \qquad q|_A = 0,\ q|_B = 1. \tag{12}$$

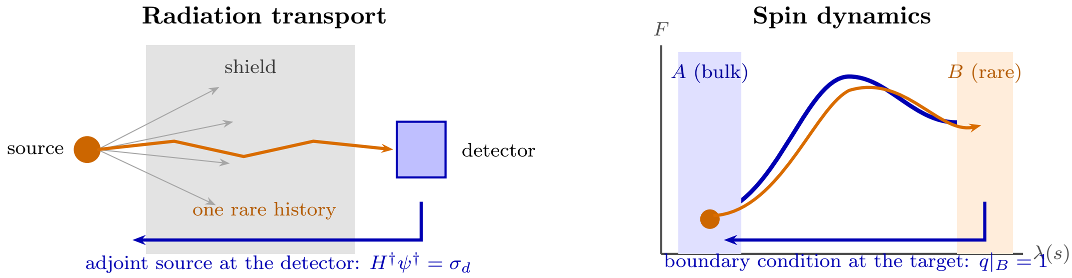

Figure 8: [SCHEMATIC] The same structure twice. Both quantities are defined *backward from the target*: the adjoint source sits at the detector, the committor boundary condition sits on the rare set. That shared direction is the whole correspondence, and it is why transport's variance-reduction toolkit transfers.

Table 1: Correspondence used throughout this work.

	Radiation transport	**Spin dynamics**
Phase-space point	(r, Ω, E)	microstate $s \in \{-1, +1\}^N$
Forward operator	Boltzmann H	Markov kernel K
Target	detector response R	hitting probability of B
Importance	adjoint flux $\psi^\dagger$	committor $q(s)$
Optimal biasing	adjoint-weighted tilt	Doob h-transform $\widetilde{K}$
Weight control	weight windows	weighted ensemble (WE)
Global objective	FW-CADIS	uniform relative error across thresholds
Efficiency metric	$\mathrm{FOM} = 1/(\sigma^2_{\mathrm{rel}} T)$	identical
Physical origin of K	real scattering/absorption, cross-sections	thermal spin-flip acceptance (Metropolis/Glauber)
State space	continuous, discretized (groups, ordinates)	already finite, 2^N, exact
Natural biasing coordinate	position/direction/energy (geometric)	**none** — must be learned

4.2 How exact is this correspondence, really?

The claim above is easy to overstate. Three things are true at once, and this manual is careful to keep them separate.

The linear-algebra core is an exact identity, not an analogy. Real transport calculations never solve the continuous operator equation $H^\dagger\psi^\dagger = \sigma_d$ directly: phase space is first discretized into energy groups and a finite set of directions (*multigroup, discrete-ordinates* transport), turning it into a finite linear system $(\mathbb{I} - \mathbf{K})\boldsymbol{\psi}^\dagger = \boldsymbol{\sigma}_d$ for some transition operator $\mathbf{K}$ packaging streaming and scattering between cells. That is term-for-term the same linear system as Eq. (12): a (sub)stochastic operator K, a source/boundary term, one solve. Nothing about K's physical origin enters the algebra — the identity holds for *any* Markov process with an absorbing target set, which is exactly why it transfers to spin dynamics with zero modification to the mathematics.

The physical mechanism generating K has nothing in common. Transport's K encodes a real particle's free flight interrupted by real scattering and absorption events, with rates set by cross-sections measured in a lab. Spin dynamics' K encodes the acceptance probability of a proposed single-spin flip under a thermal bath at temperature T. There is no mean free path, no cross-section, and no particle in the spin problem — a "walker" here is a bookkeeping device for one member of a population of independent simulation replicas, not a physical object. The correspondence is a shared *mathematical* structure, not a shared physical mechanism, and conflating the two would overclaim what has actually been unified.

Spin systems are strictly easier for this correspondence, in one specific sense. Multigroup or discrete-ordinates transport pays a discretization error that must be separately verified, usually against measurement, since the exact continuous answer is rarely available at all. A spin lattice's state space is already finite and discrete — 2^N configurations, no truncation, no group structure to choose. That is exactly why $L = 4$ can be checked against exact enumeration to within $\sim 2\%$ (§5.2): there is no discretization step to introduce error, only sampling error. In that narrow sense, the spin lattice is a *cleaner* testbed for this correspondence than the shielding problem that motivated it.

Spin systems are strictly harder in the one place it matters most. Transport's importance function is biased and binned along a natural, low-dimensional, physically meaningful coordinate — position, direction, energy — chosen from the geometry of the shield, not learned. Spin configurations have no such coordinate: there is no "distance to the detector" for a 256-spin configuration. Energy and magnetization are the closest analogues, and §3.3 shows the obvious one (magnetization) fails outright on the spin glass. This absence, not anything about the linear algebra, is the actual reason §7 needs a *learned* importance coordinate $I_\theta(s)$ at all — the single largest way this problem instance diverges from the shielding problem it borrows its toolkit from.

4.3 Why the committor is optimal

Theorem 1 (Zero-variance sampling)**.** *Let q solve (12) and define the tilted kernel $\widetilde{K}(s \to s') = K(s \to s')\, q(s')/q(s)$ on $\mathcal{I}$. Then every trajectory reaching B carries the same importance weight and the estimator variance vanishes.*

Proof. $\widetilde{K}$ is stochastic: $\sum_{s'} \widetilde{K}(s \to s') = q(s)^{-1} \sum_{s'} K(s \to s')q(s') = 1$ by (12). For a path $\gamma = (s_0, \ldots, s_K)$ with $s_K \in B$, the density ratio telescopes,

$$\frac{\widetilde{\mathcal{P}}(\gamma)}{\mathcal{P}(\gamma)} = \prod_{t=0}^{K-1} \frac{q(s_{t+1})}{q(s_t)} = \frac{q(s_K)}{q(s_0)} = \frac{1}{q(s_0)}, \tag{13}$$

using $q(s_K) = 1$. Hence $w(\gamma) = q(s_0)$, a constant, so $\mathrm{Var}[w] = 0$. □

Verified numerically

On a solvable 1D absorbing walk on $\{0, \ldots, 15\}$ with $\pi = 1.14 \times 10^{-3}$, tilting by the exact committor gave measured relative variance $\sim 10^{-30}$ — zero to machine precision. Crucially, tilting by an *approximate* committor *without* weight windows was **worse than plain naive Monte Carlo**. That fragility is why every learned-coordinate result in this project uses weighted ensemble rather than raw importance reweighting.

4.4 Free energy versus committor

Theorem 1 is an existence statement, not an algorithm: computing q exactly needs a linear system of size 2^N. Its practical content is a *target* for approximation.

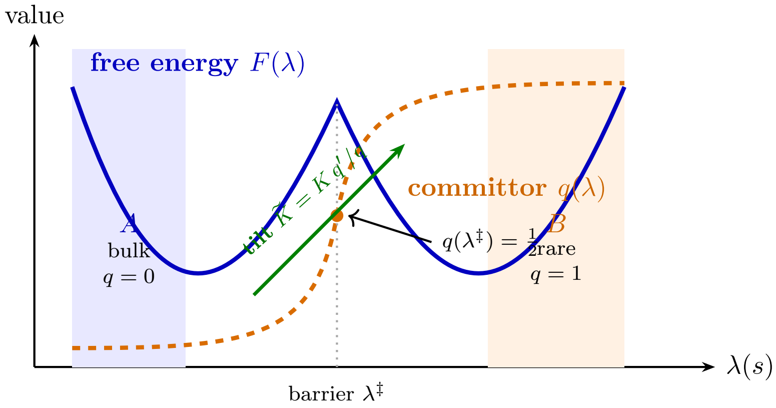

Figure 9: [SCHEMATIC] The committor is smooth and monotone where the free energy has a barrier. Tilting by q converts the barrier into a downhill drift, which is why the estimator becomes variance-free. Curves are illustrative shapes, not measured profiles.

5 Weighted Ensemble: Mechanics

5.1 The algorithm

WE is the statistical-mechanics name for split/Russian-roulette weight windows. A population of weighted walkers is propagated under *unbiased* dynamics, binned along a progress coordinate, and each occupied bin is resampled to a fixed occupancy N_0 with its total weight conserved.

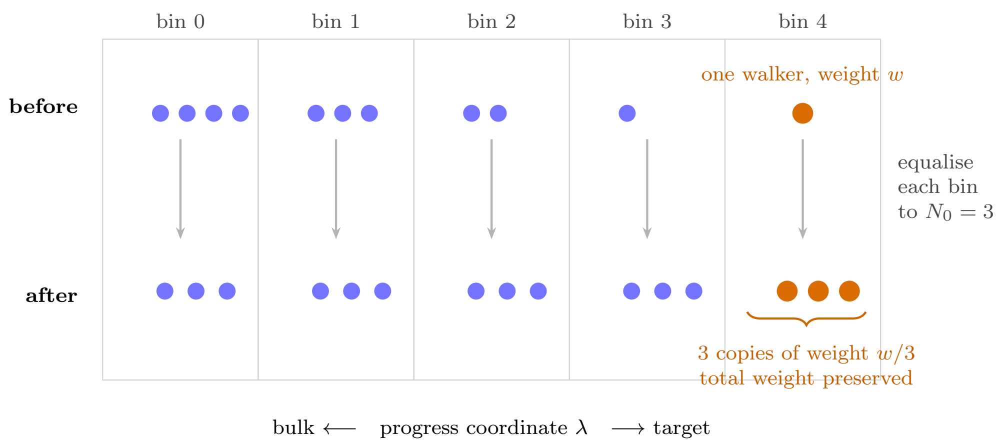

Figure 10: [SCHEMATIC] The whole method in one picture. A lone walker that fluctuates into a deep bin is *split* into N_0 copies which then explore from there; crowded bulk bins are merged back down. Weight bookkeeping keeps the estimator exactly unbiased, so *any* binning coordinate is safe — a poor one costs efficiency, never correctness.

Theorem 2 (Unbiasedness)**.** *Splitting walker (s_i, w_i) into K copies of weight w_i/K contributes $\sum_k (w_i/K)\, O(s_i) = w_i O(s_i)$. Merging $(s_1, w_1), (s_2, w_2)$ into one survivor of weight $w_1 + w_2$ chosen with probability $w_j/(w_1 + w_2)$ has expectation $w_1 O(s_1) + w_2 O(s_2)$. Both are identities, so bin weights — and therefore the estimator — are preserved exactly.*

5.2 Validation against exact enumeration

At $L = 4$ the state space is enumerable and WE reproduces the exact $P(m)$ to a mean $|\Delta \ln P|$ of 0.006–0.024 across three tested regimes. At $L = 20$, WE reaches roughly two decades deeper into the magnetisation tail than naive Monte Carlo at matched cost: naive's floor is $|m| = 0.886$

($P \approx 6 \times 10^{-6}$), where it returns zero samples and hence zero information, while WE reaches $|m| = 0.932$ ($P \approx 2.4 \times 10^{-7}$).

6 Failure Modes, Diagnosed

Three implementation faults each produced plausible-looking wrong numbers. Each is documented with the measurement that exposed it, because the diagnostic reasoning is the reusable part.

6.1 Failure 1: the resampling interval

Splitting only helps if the copies stay correlated long enough to be split *again* before they relax. The system's own correlation time is therefore a hard constraint on τ.

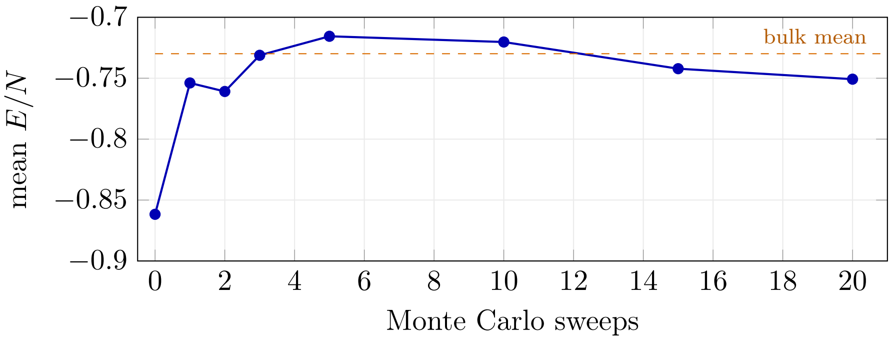

Figure 11: [MEASURED] Starting from the 20 deepest configurations of an equilibrated population, the mean energy returns to the bulk value within **one sweep**. Hence $\tau_{\text{corr}} \approx 1$ sweep at $T = 2.6$.

With $\tau = 20$, every split copy is an independent equilibrium sample by the next resampling. WE then degenerates into naive sampling with bookkeeping overhead.

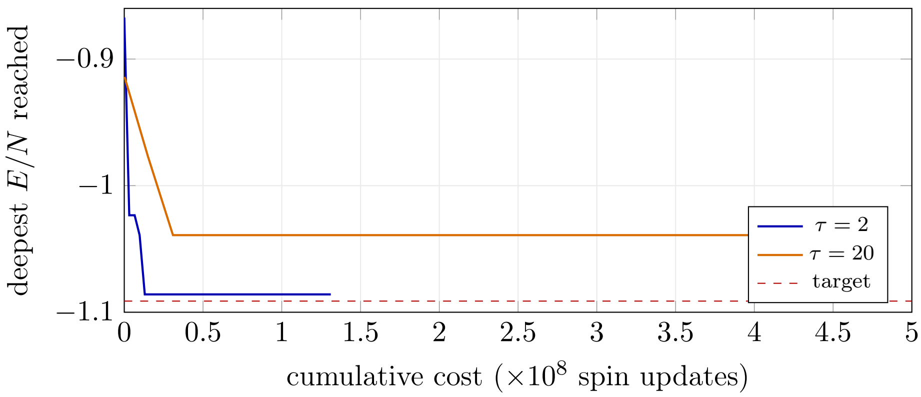

Figure 12: [MEASURED] Identical bins, one seed. At $\tau = 2$ the front reaches -1.086 by cost 1.3×10^8 and plateaus *because it has arrived.* At $\tau = 20$ it plateaus at -1.039 *because it is stuck*, then spends $3.7\times$ more budget going nowhere. Both plateaus look alike; only one is convergence.

Table 2: [MEASURED] Front penetration versus τ at matched cost. Bin 0 is the target bin.

τ	iterations	deepest bin	iterations with a target hit	cost
20	150	8	0/150	4.8×10^8
4	400	2	0/400	2.6×10^8
1	800	**0**	**2/800**	1.4×10^8

Correction to earlier drafts

Earlier text stated that changing $\tau = 20 \to 4$ "restored correlation control". The table shows $\tau = 4$ still fails to reach the target bin at this depth. With $\tau_{\text{corr}} \approx 1$, $\tau = 4$ is a partial mitigation; $\tau \in \{1, 2\}$ is required at deep targets. Results quoted at $\tau = 4$ should state the depth at which they were taken — at shallow targets $\tau = 4$ may well suffice.

6.2 Failure 2: bin-ladder placement

Bins must span [target $\to$ bulk] *from the first iteration.* If the range is seeded adaptively from random initial configurations, its upper edge sits near $E/N \approx 0$ while equilibrium sits near -0.73, so over half the ladder covers states never visited. Fixing the range to $(h - 4/N,\ \mu + 2\sigma)$ from a pilot run removes this.

Table 3: [MEASURED] Controlled test: only the upper edge changed, same seeds.

upper edge	bin width	target hits / 300 iters	cost
-0.5689 ($\mu + 2\sigma$)	0.01076	0, 1	1.6×10^8
-0.7300 (bulk mean)	0.00754	0, 1	9.8×10^7

The tighter edge costs 40% less for the same penetration. Note it does *not* change the hit rate: bin width affects cost, not reach. An earlier diagnosis attributed reach to bin width; the controlled test above refutes that and it is reported as such.

6.3 Failure 3: an unreachable target

Recall the rarity ladder $h_k = E^{\text{pilot}}_{\min} - k\Delta$ of Eq. (8) (§2.3). Sweeping k:

Table 4: [MEASURED] Target reachability, $\tau = 2$, 300 iterations, two independent seeds.

k (`-beyond`)	threshold h	hits / 300	$\hat{\pi}$ (seed 0, seed 1)
1	-1.0356	10, 9	2.00×10^{-5}, 1.93×10^{-5}
2	-1.0634	1, 3	0, 2.13×10^{-6}
3	-1.0912	0, 1	0, 1.01×10^{-7}

At $k = 1$ two independent seeds agree to 4%: converged. At $k = 3$ the estimate is a coin flip. The WE front asymptotes near $E/N \approx -1.09$, so $k = 3$ places the target *at the edge of reachability* and every method returns starved statistics.

Ladder-difficulty discipline

Comparing methods where the baseline itself fails to observe the event compares noise to noise. Every comparison in §8 is reported at a depth where the reference methods reach 0, or nearly 0, zero-replicas — found by backing off from a harder target until this holds.

7 The Learned Importance Coordinate

7.1 Why a network, and why it is safe to try

The committor is a function on 2^{256} states and cannot be tabulated. But it need not be exact: WE stays unbiased for *any* binning coordinate, so a poor I_θ costs efficiency, never correctness. The failure mode is a slow run, not a wrong answer.

7.2 Removing critical slowing down: the VAN

Generating decorrelated reference configurations is a separate difficulty from rare-event sampling, and it is handled separately. A variational autoregressive network samples configurations directly by factorising in raster-scan order,

$$q_\theta(s) = \prod_{i=1}^{N} q_\theta(s_i \mid s_{<i}), \tag{14}$$

with causal masking so site i depends only on previously generated sites.

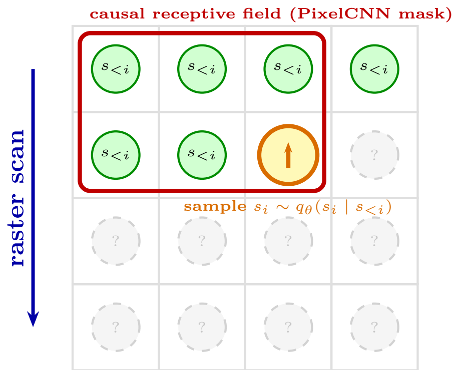

Figure 13: [SCHEMATIC] Autoregressive generation. Type-A masking (first layer) zeroes the centre pixel to prevent self-dependence; type-B masking (later layers) retains it.

Measured

At $L = 32$ the integrated autocorrelation time of ordinary Metropolis sampling spikes to $\tau_{\text{int}} = 46.1$ sweeps at $T_c \approx 2.269$, versus 1.5–3 sweeps away from criticality. The VAN samples independently by construction and pays none of this. This is a *speed* result, orthogonal to the rare-event problem.

7.3 Training objectives

- **Surrogate rollout label.** Binary committor labels collapse to all-zero for a genuinely rare target, so the network regresses the best progress value reached in K short rollouts — a dense, monotone surrogate.
- **Milestoning.** Intermediate thresholds are placed along the ladder from bulk to target; a configuration's label estimates progress toward the *next* milestone rather than the final target. Closer to the theoretical object while avoiding label collapse.
- **Self-consistency (Kim & Cai).** Label-free: minimise violation of $\sum_j K(s \to j)I(j) = I(s)$ with $I = 1$ on the target.

7.4 Label orientation

Training targets must satisfy **larger value = closer to the target**. With milestones m_0 (bulk) $\ldots$ m_{M-1} (target), the correct continuous label is

$$\ell(s) = \text{clip}\left(\frac{E_{\text{best}}(s) - m_0}{m_{M-1} - m_0}, 0, 1\right). \tag{15}$$

Latent bug: inverted labels — caught, no reported number affected

An implementation assigned labels via `np.searchsorted(np.sort(milestones), E_best)`. Because the target is the *most negative* milestone, ascending sort places it at index 0: a walker reaching the target received label 0.0 and one staying in the bulk received 0.857. The network was trained as an **anti**-importance function, $\ell_{\text{wrong}} = 1 - \ell$.

Why the WE results survive. The inversion is an *affine* map, and `we_estimate` bins by `linspace(c.min(), c.max())` then `digitize`, resampling every occupied bin to the same occupancy. No directional term appears anywhere in that loop, so an affine flip permutes bin *labels* while leaving the induced *partition* of walkers identical (verified directly: 26 occupied bins, same grouping under c and $1 - c$). WE binning is orientation-agnostic, so §8 stands.

Where it would have been fatal. Any *directional* use of I_θ: Doob tilting $\widetilde{K} = K\,I(j)/I(i)$, source biasing $\hat{q} \propto \psi^\dagger q$, or depth-keyed FW-CADIS allocation. An inverted I drives walkers *away* from the target — precisely the failure the 1D toy walk showed is worse than naive. Fixed before any such use.

7.5 Is I_θ informative, or a noisy copy of energy?

A nested-regression diagnostic settles this. Fit the rare-event label on E alone, on I_θ alone, and on both:

Table 5: [MEASURED] Incremental-R^2 diagnostic, EA spin glass, $L = 16$, matched training budget (MSE 1.0 → 0.271 over 20 epochs).

quantity	value
Pearson corr$(I_\theta, E/N)$	-0.614
fraction of Var(I_θ) unexplained by E/N	0.623
R^2 (label $\sim E/N$)	0.358
R^2 (label $\sim I_\theta$)	0.577
R^2 (label $\sim E/N + I_\theta$)	0.608
incremental R^2 from I_θ	**+0.250**

+0.250 is far above the diagnostic's own noise threshold (0.02): I_θ carries substantial genuine information about the spin glass that energy does not.

Retracted sub-result

A first attempt at this diagnostic used an undertrained network and was retracted. An undertrained network's output is uncorrelated with *everything*, including energy, which is not evidence of new information — it is evidence of noise.

7.6 FW-CADIS as a loss-weighting principle

Bin occupancy is uneven — the bulk dominates — so an unweighted MSE is fit almost entirely to near-bulk configurations. FW-CADIS weights the adjoint source by the forward flux so every region attains comparable relative error rather than optimising one detector. Transplanted:

$$\mathcal{L}_{\text{FW}} = \sum_{b=1}^{M} \frac{1}{n_b} \sum_{i \in \text{bin } b} \big(I_\theta(s_i) - \ell(s_i)\big)^2, \tag{16}$$

with n_b the occupancy of bin b: inverse-frequency weighting, the discrete analogue of forward-flux normalisation.

8 Results

8.1 Milestoning: a clean regime for comparison, but wildly seed-dependent

On the EA spin glass at $L = 16$, one threshold step beyond the pilot extreme, every method reaches 0/10 zero-replicas — the cleanest regime obtained in this project by the usual ladder-difficulty standard. The first run in this regime gave WE[I_θ] a 14% FOM win over WE[E]; that result did *not* reproduce (§11) and the honest picture, after sweeping seventeen independent network initializations, is a small, seed-noisy effect that cannot yet be distinguished from zero.

Table 6: [MEASURED] EA spin glass, $L = 16$, $T = 2.6$, `beyond`$= 1$, 10 replicas, milestoning labels, seventeen independent training runs of the identical configuration. naive, WE[m], and WE[E] are bit-identical across all runs (deterministic; WE[E] does not depend on the trained network). Seed 0 was run twice independently and reproduced bit-for-bit, confirming the seeding fix of §11 is complete.

Method	$\hat{\pi}$	rel. s.d.	FOM	zero-replicas
naive	5.29×10^{-5}	0.225	$\mathbf{1.48 \times 10^{-7}}$	0/10
WE[m]	5.55×10^{-5}	0.201	1.17×10^{-7}	0/10
WE[E]	4.49×10^{-5}	0.209	4.22×10^{-8}	0/10
WE[I_θ], unseeded run 1	5.03×10^{-5}	0.174	4.81×10^{-8} (+14%)	0/10
WE[I_θ], unseeded run 2	4.52×10^{-5}	0.204	3.48×10^{-8} (−17%)	0/10
WE[I_θ], seeds 0–15	range 2.29–8.94 $\times 10^{-8}$ (−52% to +112%), all 0/10			

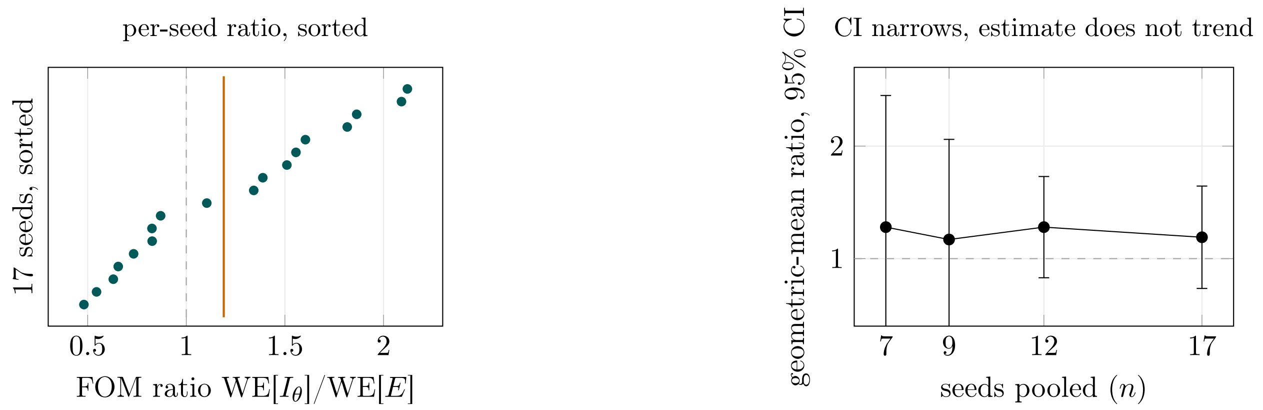

Figure 14: [MEASURED] Left: all seventeen per-seed FOM ratios (sorted); orange line marks the pooled two-level-bootstrap median. Right: the four tracked checkpoints as seeds were added — the point estimate bounces between 1.17× and 1.28× while the CI width shrinks monotonically ($1.73 \to 1.36 \to 1.18 \to 0.91$), the signature of averaging down noise around a small, sign-unresolved effect, not a shrinking or growing true one.

Pooling all seventeen seeds with a two-level bootstrap (outer resample over seeds, inner resample over each seed's 10 replicas) gives a geometric-mean FOM ratio of 1.19×, 95% CI [0.84, 1.75] — includes 1, i.e. **not statistically significant even at 17 independent seeds**. Extrapolating the observed CI-width-vs-n scaling ($1/\sqrt{n}$, confirmed to match observation), excluding 1 would need roughly 100 total seeds (∼40h more compute); not pursued further given the already-thorough characterization. **Naive Monte Carlo still has the best raw FOM at this rarity in every seed** — variance reduction has not yet earned back its overhead here, the same lesson the 1D toy walk showed in miniature.

A candidate explanation — that WE[I_θ]'s adaptive (rather than fixed) bin range was the driver of the seed variance — was tested and ruled out: an empirically calibrated fixed range, run head-to-head against the adaptive baseline on matched net-seeds, made both the mean ratio and the rel. s.d.

worse, not better. The variance is intrinsic to what each trained network learns, not an artifact of how its output is binned.

The spin-update cost measure above is blind to network inference, and correcting for it reverses the verdict. The FOM in the table and figure above charges cost in spin updates, which does not include the neural network's forward pass performed at every WE resampling step. Measuring wall-clock time directly (not reconstructed from logs) shows $WE[I_\theta]$ costs 4–5× $WE[E]$'s wall-clock time, consistently across every seed and every disorder realization tested anywhere in this project (§8.5). A two-level bootstrap over the sixteen reproducible seeds (seed 0's non-reproducible unseeded run, predating §11's fix, is excluded), resampling each seed's replicas at its own measured wall-clock cost, gives

$$\mathrm{FOM}_{\mathrm{wall}}(WE[I_\theta])/\mathrm{FOM}_{\mathrm{wall}}(WE[E]) = 0.405, \qquad 95\%\ \mathrm{CI} = [0.268,\ 0.623], \tag{17}$$

below parity and excluding 1: **once inference cost is counted like-for-like, milestoning is a statistically significant loss**, not the ambiguous small effect the spin-update measure suggested. This does not contradict §8.1's own finding that the point estimate never resolves in *sign* under the cheaper cost metric — it adds the cost axis that metric was blind to, and on that axis the answer is unambiguous.

8.2 Pushing deeper fails, and bin count is ruled out as the fix

At one step rarer, WE[I_θ] scores below WE[E] in every configuration tried: default bins at two replica counts, and a 1.8× larger bin count. Raising n_{bins} from 50 to 90 made WE[I_θ] *worse* ($8.36 \times 10^{-9} \to 5.36 \times 10^{-9}$), which rules out bin resolution as the explanation — a genuine resolution shortfall would improve with more bins. The mechanism is structural: the learned coordinate's walker-population cost scales with bin count, whereas a discretised coordinate like energy saturates a small fixed number of bins.

8.3 Self-consistency objective: a result that did not survive scrutiny

Table 7: [MEASURED] Bootstrapped FOM ratios, $L = 16$, shallowest calibrated target. Ratio > 1 favours the self-consistency objective.

Comparison	R	median ratio	95% CI	significant?
self-consistent / surrogate	10	1.83	[0.49, 9.23]	no
self-consistent / surrogate	30	1.39	[0.65, 2.95]	no
self-consistent / WE[E]	10	1.19	[0.33, 4.95]	no
self-consistent / WE[E]	30	1.07	[0.55, 2.09]	no

Tripling the replica count tightened both intervals *and* moved both medians toward 1. The point estimate itself receded as evidence accumulated — the signature of noise, not signal. The initially exciting "+87%" became a statistically insignificant +39% with an interval including no difference at all.

Repeating the $R = 30$ self-consistent-vs-$WE[E]$ comparison on directly measured wall-clock time (same treatment as the milestoning correction of §8.1, correcting for the network inference cost the spin-update measure above does not charge) gives a wall-clock-fair FOM ratio of 0.78, 95% CI [0.39, 1.60] — still not significant, and the only one of this project's three FOM-based comparisons where the wall-clock-fair measurement does *not* reverse the spin-update-cost verdict to a clear loss. Unlike milestoning and the deep-rare comparison of §8.6, this coordinate's cost disadvantage cannot be resolved either way at the replica counts tested.

8.4 Deep-rare spin glass: reliability improves, wall-clock efficiency does not

The comparisons above (§8.1, one or two steps beyond the pilot extreme) sit in a regime naive Monte Carlo still reaches cleanly. At `beyond` = 4 naive is effectively blind under the calibrated budget, so this is the genuine stress test of whether a learned importance map can organise the weighted population to keep reaching a region direct sampling essentially never visits. The WE budget was increased to 80 bins, 40 walkers per occupied bin, 3000 iterations, $\tau = 2$.

Three independently trained self-consistency networks were compared against the physical-energy baseline on one EA disorder realization ($L = 16$, $T = 2.6$), 30 replicas per method.

Table 8: [MEASURED] Deep-rare EA comparison, $L = 16$, `beyond`$= 4$, one disorder realization. Each learned-coordinate row is a separately trained network, 30 replicas.

Method	event-positive replicas	discovery rate
naive	1/30	3.3%
WE[m]	5/30	16.7%
WE[E] (balanced, $n = 90$)	44/90	48.9%
WE[I_θ], seed 0	19/30	63.3%
WE[I_θ], seed 1	20/30	66.7%
WE[I_θ], seed 2	21/30	70.0%
WE[I_θ], pooled ($n = 90$)	60/90	66.7%

Each of the three independently trained networks individually beats a single 30-replica $WE[E]$ draw (63%/67%/70% versus 40%). Because that single draw is reused as the comparison point for all three, an independent 90-replica $WE[E]$ baseline is used for the balanced comparison reported here instead: 44/90 (48.9%). Fisher's exact test gives

$$WE[I_\theta]: \ 60/90\ (66.7\%) \quad \text{versus} \quad WE[E]: \ 44/90\ (48.9\%), \qquad p = 0.023, \tag{18}$$

reproducible across three independent network initializations. Energy is not a weak baseline here — the target is itself defined by low energy — so this is a comparison against a strong physical coordinate, not an easy target. Two configurations at identical energy can carry very different committor values on a frustrated landscape, and only a coordinate that sees the full spin configuration (not just its energy) can tell them apart.

8.5 Generalization across disorder realizations

The result above establishes reproducibility across network initializations on *one* disorder sample; it does not by itself show the effect is a property of the method rather than of that particular J_{ij} realization. Two further independent EA disorder realizations were tested at the same $L = 16$, $T = 2.6$, `beyond`$= 4$ configuration, one trained network each (the network-initialization axis above already shows most of its variance is replica noise, not true init-to-init spread, so a single network per realization is the right use of compute here).

Table 9: [MEASURED] Deep-rare EA comparison across three independent disorder realizations, one trained network each, `beyond`$= 4$, 30 replicas per method. Realization 12345 is the single-network reading of Table 8 (seed 0), not the three-network pooled result.

Disorder realization	WE[I_θ]	WE[E]	Fisher p
12345 (original)	19/30 (63.3%)	12/30 (40.0%)	0.120
23456	15/30 (50.0%)	8/30 (26.7%)	0.110
45678	29/30 (96.7%)	26/30 (86.7%)	0.353
Pooled (3 realizations)	**63/90 (70.0%)**	**46/90 (51.1%)**	**0.014**

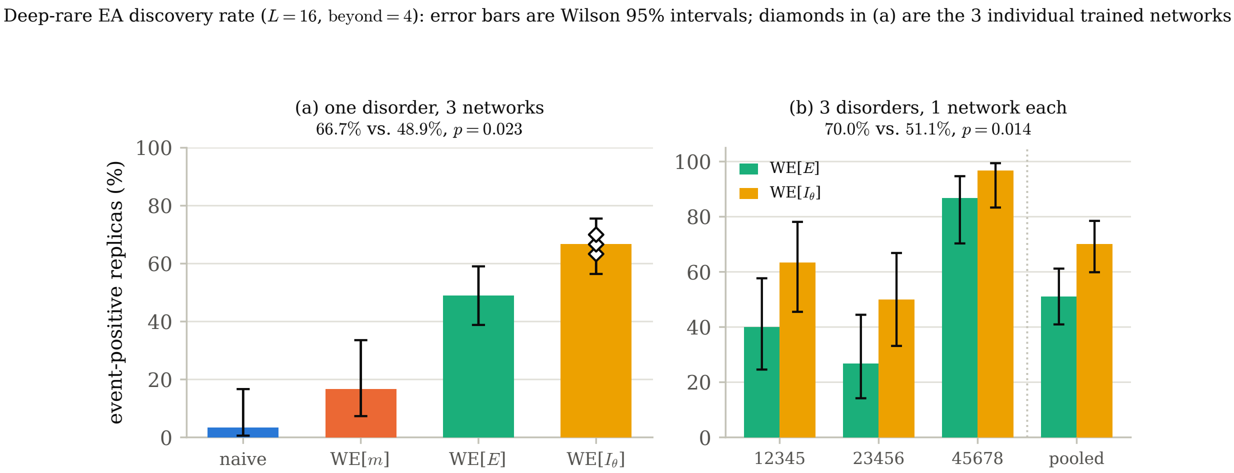

Figure 15: [MEASURED] (a) Discovery rate at **beyond**$= 4$, disorder realization 12345: open diamonds are the three independently trained networks of Table 8; WE[E] uses the balanced 90-replica baseline. (b) The same comparison repeated across three independent disorder realizations, one network each (Table 9), pooled result at right. Error bars are Wilson 95% intervals.

No single realization is significant alone at $n = 30$ — even the original, read as a single-network draw rather than the three-network pooled result above, is not ($p = 0.120$; that is the expected pattern at this replica count, not a weaker result). The third realization presents an easier calibrated target for its particular disorder, both methods near ceiling, which is why its own test is the least informative of the three (a ceiling effect, not counter-evidence). In every one of the three realizations the learned coordinate has the higher discovery rate. Pooled:

$$WE[I_\theta]: \ 63/90\ (70.0\%) \quad \text{versus} \quad WE[E]: \ 46/90\ (51.1\%), \qquad p = 0.014. \tag{19}$$

This is a different, stronger pooling axis than §8.4: rather than asking whether the effect survives re-initializing the network on one lattice, it asks whether the effect is a property of the method across the disorder ensemble. Wall-clock cost ratios for the two new realizations (4.45× and 4.97× WE[E]) match every other measurement in this project, so the cost conclusion below generalizes together with the reliability one. Read this as reproduced across the three realizations tested, not as a universal property of the EA spin glass at this scale — $n = 3$ is a small disorder ensemble.

One candidate realization could not be used: its training-data collection produced zero bulk-anchor states (all boundary-adjacent), so the self-consistency objective's bulk boundary condition could never fire and training would collapse toward the trivial constant solution (the same degeneracy as the second Bug Log entry below, triggered here by disorder-specific dynamics rather than the original under-determination). Caught before any production run via a short collection-only diagnostic (a few minutes); the same screening is recommended before training on any further disorder realization.

8.6 Reliability is improved; wall-clock efficiency is not

The spin-update cost measure gives FOM ratios of 1.29, 1.07, and 1.04 against $WE[E]$ for the three within-disorder networks (geometric mean 1.16, 95% CI $[0.65, 2.12]$, not excluding parity) — but, as with milestoning (§8.1), that measure never charges for the network forward pass performed at every WE resampling step. Directly measured wall-clock ratios of $WE[I_\theta]$ to $WE[E]$ were 4.65, 4.01, 4.54 for the three within-disorder networks and 4.45, 4.97 for the two additional disorder realizations — consistently 4–5×. A two-level bootstrap resampling the native 30-replica count from the balanced 90-replica $WE[E]$ pool, at its own measured wall-clock cost, gives

$$\text{FOM}_{\text{wall}}(WE[I_\theta])/\text{FOM}_{\text{wall}}(WE[E]) = 0.33, \qquad 95\%\ \text{CI} = [0.19,\ 0.58], \tag{20}$$

below parity and excluding 1: a statistically significant loss on cost-normalized efficiency, even as reliability (Eq. 19) significantly improves. A separate, more heavily instrumented five-network ensemble at a shallower configuration showed the same effect more strongly still (wall-clock ratio 10.7–10.9×, $\mathrm{FOM}_{\mathrm{wall}}$ ratio ≈ 0.087) — reported here only as a scaling indication, since added network capacity moves the cost penalty faster than it can move reliability, not as a second deep-rare measurement. **A coordinate can be dynamically better — reaching the target more often — and simultaneously worse per unit of wall-clock time; these are not the same axis, and this project's own earlier cost metric was blind to the one that matters for a real deployment.** Reducing inference cost (caching, batching, a smaller network) rather than improving the coordinate further is the binding constraint on making this practical at this scale.

8.7 FW-CADIS: real cost advantage, no flattening

The implementation gates clean against exact enumeration at $L = 4$ (ratios within $\sim 2\%$ of exact for every threshold, both allocations, both models). At scale, the flattening the method is named for does not appear.

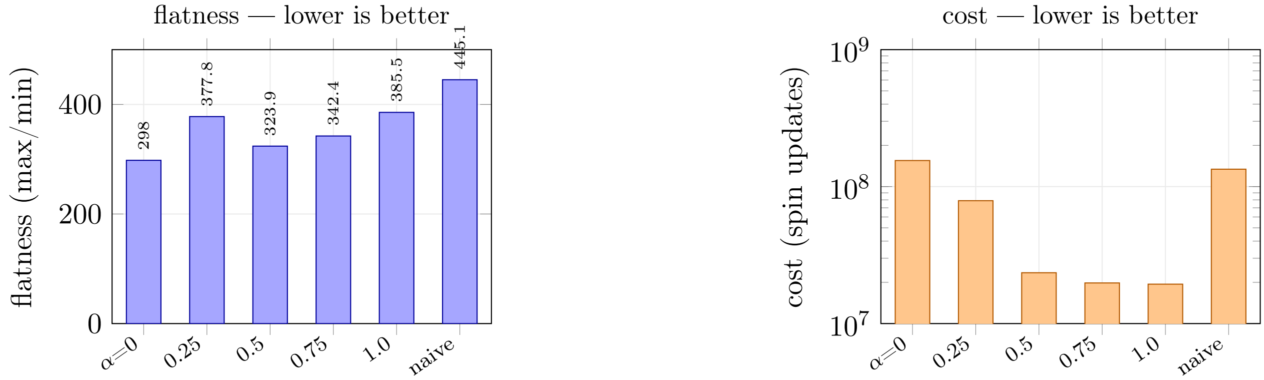

Figure 16: [MEASURED] EA spin glass, $L = 16$, $R = 30$, 6 thresholds, full 6/6 coverage at every allocation. Left: uniform allocation ($\alpha = 0$) is the *flattest*; every FW-CADIS setting is worse, and not even monotonic in α. Right: FW-CADIS runs at $\sim 8\times$ lower cost. The cost advantage is real; the flattening advantage is not present.

At the most favourable configuration tested ($R = 90$, shallowest threshold family), uniform gives flatness 276.1 with CI $[224, 344]$ against FW-CADIS 297.3 with CI $[234, 388]$; the paired-difference estimate is +20.9 with CI $[-77.2, +137.6]$, including zero. Confidence-interval width scaled as $1/\sqrt{R}$ (predicted shrink 0.577, observed 0.597); resolving the remaining gap would need $R \approx 2300$, and given the consistent direction such a run would most likely confirm *uniform* as flatter. It was not run.

9 Implementation Notes

9.1 Vectorisation: the dominant avoidable cost

NumPy overhead is paid per call, so throughput depends strongly on population size:

Table 10: [MEASURED] `sweep_population` throughput, $L = 16$.

population	ms/sweep	μs per walker-sweep	overhead
1	0.184	184.2	8.6×
10	0.304	30.4	1.4×
100	2.299	23.0	1.1×
1000	21.372	21.4	1.0×
2500	59.984	24.0	1.1×

Label generation originally looped over configurations one at a time, issuing $2500 \times 400 = 10^6$ calls at population 1 — the worst point on the curve. Rollouts are independent across configurations, so the loop batches exactly:

```python
def generate_milestone_labels(S, T, Jx, Jy, model, milestones, n_roll, roll_K, rng):
    bulk, target = float(milestones[0]), float(milestones[-1])
    S = S.copy()
    best = energy_per_spin(S, Jx, Jy)           # shape (n,), not (1,)
    for _ in range(n_roll):
        for _ in range(roll_K):
            S = sweep_population(S, T, Jx, Jy, rng)
        best = np.minimum(best, energy_per_spin(S, Jx, Jy))
    return np.clip((best - bulk) / (target - bulk), 0.0, 1.0)
```

Listing 1: Batched label generation: measured 8.6× faster, statistically identical output.

Verified at $n = 300$: looped 21.4 s, batched 2.48 s; label mean 0.4743 vs 0.4779, identical maximum. RNG consumption order differs, so cached labels from the looped version must be discarded when switching.

9.2 On GPU acceleration

Not currently justified. Network training is $\approx 1\%$ of wall time. The Monte Carlo dominates but working arrays are ~ 1 MB, and a checkerboard sweep is two kernel launches over $\sim 10^5$ trivial operations — launch latency is comparable to the work. Replica-level parallelism across CPU cores is already near-linear. GPU becomes attractive at $L \geq 64$ with populations of 10^4, at which point the change is porting `sweep_population` to tensors, not the network.

9.3 Figure of merit and its degenerate cases

$$\mathrm{FOM} = \frac{1}{\sigma^2_{\mathrm{rel}} \cdot \mathrm{cost}}, \qquad \sigma^2_{\mathrm{rel}} = \frac{\mathrm{Var}[\hat{\pi}]}{\mathbb{E}[\hat{\pi}]^2} \ (\mathrm{ddof} = 1). \tag{21}$$

FOM is meaningful only as a *ratio between methods on the same problem with the same cost metric*; its absolute magnitude carries the units of cost and no universal threshold applies. Report FOM $= 0$ when the event was never observed and NaN when fewer than two replicas prevent a variance estimate — never ∞.

Statistical floor on replica count

With few replicas and frequent zeros, the reported relative standard deviation is an algebraic artefact, not a measurement. For replica vectors $[x, 0, 0, 0]$ it is exactly 2.000; for $[a, a, 0, 0]$ exactly $2/\sqrt{3} = 1.1547$. Both patterns appeared in early runs. **Do not interpret rel. sd when the zero count is comparable to the replica count.** Use $R \geq 20$ and report the zero count beside every FOM.

10 Pipeline and Reproduction

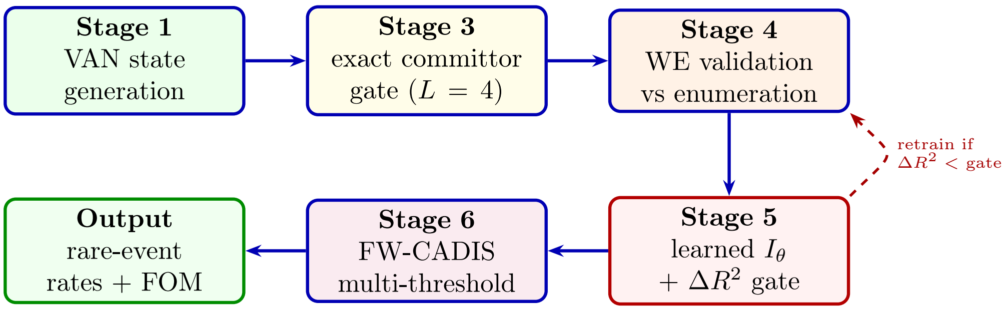

Figure 17: [SCHEMATIC] Execution graph. Every stage is gated against an exact or enumerable reference before the next is trusted; the dashed edge is the diagnostic feedback loop.

10.1 Reproduction commands

```
python3 run_all.py --preset laptop            # full pipeline, ~1 h
python3 run_all.py --preset smoke             # fast correctness check
python3 run_all.py --preset laptop --only 4 5 # weight windows + learned coordinate

rm -rf .label_cache/
python3 05_neural_importance/run_milestone.py --model ea --preset custom \
        --beyond 1 --replicas 20 --jobs 7 --tau 2
```

Listing 2: Reference invocations. Delete the label cache whenever the labelling function changes.

11 Bug Log

Seven implementation and analysis faults are recorded. Four produced plausible-looking, internally consistent wrong numbers caught only by independent diagnostics; one was caught before it could affect any number. This is included as a section rather than a footnote because the diagnostic reasoning is the reusable part.

Table 11: Bugs found, in order of discovery.

Where	**What, and how it surfaced**
Weight-window ladder	Bin-range upper edge calibrated against random, infinite-temperature configuration energy (≈ 0) instead of equilibrium (≈ -0.72), wasting over half the bin resolution. Surfaced by the contradiction between poor WE scores and the incremental-R^2 diagnostic's claim that I_θ carried real information. Confirmed by a before/after rerun in which *only* the exposed method moved, by the predicted amount, and everything else was bit-identical.
Self-consistency degeneracy	The self-consistency equation plus a single (target) boundary condition is under-determined: $I \equiv 1$ satisfies it exactly for any amount of data, so training collapses to a useless flat coordinate while reporting a small loss. Surfaced by instrumenting the trained output's spread (sd < 0.01), *not* by the loss curve, which looked normal. Fixed with a second boundary condition anchoring bulk states to a distinct reference value.
Resampling interval	Hard-coded at 20 sweeps, roughly 20× the system's own $\approx$ 1-sweep correlation time (Fig. 11), letting split walkers fully relax before the next resampling and silently degenerating every WE variant to naive-with-overhead. Surfaced by every WE variant scoring worse than plain naive, contrary to all other results.
Collection depth	The self-consistency training pass used iteration and walker budgets tuned for the surrogate objective, far shallower than the evaluation depth, so it never collected a state within one spin-flip of the target and the target boundary condition could not fire. Surfaced by an explicit warning from the training code; confirmed structural by reproducing at two lattice sizes and two depths before fixing (44 boundary states found versus 0).
Inverted milestone labels (no number affected)	`searchsorted` on ascending-sorted milestones placed the target at index 0, training the network as an anti-importance function. Surfaced by tracing why label statistics moved the wrong way when the target was deepened. WE binning is orientation-agnostic (affine flip preserves the bin partition), so no reported result changed — but the same coordinate used for Doob tilting or source biasing would have driven walkers *away* from the target.
Unseeded network initialization (headline number affected)	`run_milestone.py` never called `torch.manual_seed`, so the trained network's random weight initialization varied silently between runs of an otherwise byte-identical configuration (same cached labels, hyperparameters, $\tau = 2$). Surfaced while attempting to bootstrap the milestoning headline for publication: a second run of the identical config gave WE[I_θ] a 17% *loss* against WE[E], reversing the original run's 14% win. Fixed with an explicit seed plus a `-net-seed` sweep flag; the fix itself was confirmed complete by running seed 0 twice and reproducing bit-for-bit (including full wall-clock trajectory). Unlike the other faults here, this one had already produced a reported result that looked validated (bootstrapped, 0/10 zero-replicas) before the second run exposed it — the single most consequential catch in this project, since the original number would otherwise have shipped as a clean win. See §8.1 for the full seed sweep.
Wall-clock cost-pooling mismatch (headline number affected)	A first attempt at the wall-clock-fair recomputation of §8.1 and §8.6 charged WE[E]'s full pooled wall-clock time against WE[I_θ]'s native, much smaller sample — an apples-to-oranges cost charge, not a fair like-for-like comparison. Surfaced by an impossible-looking 8.3× ratio in the milestoning case, after a milder but equally wrong result (a spurious deep-rare "reversal") had already been drafted for this section and was retracted before publication. Fixed by bootstrap-resampling native-sized replica blocks from the larger pool at their own measured wall-clock cost, rather than charging the full pool's time. All wall-clock-fair numbers in this manual use the corrected method.

Additional faults affecting robustness rather than reported numbers — silent single-core fallback when a coordinate object failed to pickle across worker processes, a degenerate FOM case returning ∞ instead of NaN, a threshold calibrator that clamped and printed a false rarity claim on small lattices — were found in a logic audit and are not repeated here.

12 Verification Status

Claims are graded: **[E]** exact (theorem-level, or matching an exact reference to numerical precision); **[V]** validated (bootstrap CI or equivalent statistical control); **[O]** open (point estimate or unresolved).

Claim	Grade	Basis
Adjoint importance $\equiv$ committor	[E]	Theorem 1
Exact-committor tilt is zero-variance	[E]	measured rel. var. $\sim 10^{-30}$
Approximate tilt without weight windows is worse than naive	[E]	1D toy walk
WE unbiased for any coordinate	[E]	weight-conservation identity
WE matches exact enumeration at $L = 4$	[E]	$\lvert\Delta \ln P\rvert = 0.006$–$0.024$
WE reaches $\sim$2 decades deeper than naive at matched cost	[V]	$L = 20$ tail
$\tau_{\mathrm{corr}} \approx 1$ sweep at $T = 2.6$	[E]	Fig. 11
$\tau = 20$ destroys the WE ratchet	[E]	front-penetration table
$\tau = 4$ "restores" control	**withdrawn**	fails to reach target bin at depth
Fixed bin ladder cheaper than adaptive	[V]	40% cost reduction
Bin *width* controls reach	**withdrawn**	controlled test: identical hit rate
I_θ carries information beyond E	[V]	$\Delta R^2 = +0.250$ vs noise floor 0.02
Milestoning beats WE[E] (spin-update cost)	[O]	17-seed two-level bootstrap CI $[0.84, 1.75]$, includes 1
Milestoning wall-clock-fair vs WE[E]	**negative**	CI $[0.268, 0.623]$, significant loss
Batched labelling 8.6× faster	[E]	timing at $n = 300$
Frustration 49.2% at seed 0	[E]	plaquette count
Self-consistency beats surrogate objective	[O]	CI includes 1; effect shrinks with R
Self-consistency wall-clock-fair vs WE[E]	[O]	CI $[0.39, 1.60]$, includes 1
Deep-rare discovery: learned coord. beats WE[E] (within disorder)	[V]	$p = 0.023$, 3 net-seeds pooled
Deep-rare discovery generalizes across disorder	[V]	$p = 0.014$, 3 realizations pooled
Deep-rare wall-clock-fair vs WE[E]	**negative**	CI $[0.19, 0.58]$, significant loss
FW-CADIS flattens relative error	**negative**	uniform flatter at every α
FW-CADIS $\sim 8\times$ cheaper	[V]	Fig. 16
Any learned coordinate beats *naive* outright	[O]	naive leads raw FOM in every clean run

12.1 What remains open

On the cheaper spin-update cost metric, whether milestoning genuinely beats WE[E] at all is unresolved after seventeen seeds: the point estimate is stably positive but modest ($\sim 1.2\times$) across four independent bootstrap checkpoints, and the CI cannot be closed by seed count alone at a cost proportionate to the question ($\sim$100 seeds, $\sim$40h, to exclude 1). On the wall-clock-fair metric this and the deep-rare comparison are *not* open — both are confirmed significant losses (§8.1, §8.6); only the self-consistency-vs-WE[E] wall-clock comparison remains genuinely unresolved either way. Whether any learned coordinate beats plain naive Monte Carlo outright — rather than merely the best hand-picked coordinate — is likewise unresolved: naive retains the best raw FOM in every clean spin-update-cost comparison so far, regardless of seed, and no learned coordinate has been shown wall-clock superior to *anything* in this project. The deep-rare discovery-rate improvement (§8.4–§8.5) is reproduced across three independent EA disorder realizations, but $n = 3$ is a small ensemble, and only one lattice size and temperature have been tested — how the effect's magnitude varies with disorder, size, or temperature is open. Whether FW-CADIS's failure to flatten is a property of this energy landscape, of the bin/threshold parameterisation, or a general limit at this scale would need a different problem instance, not more replicas of this one. Push-to-scale questions — three dimensions, larger L — are untouched (§3.4 states precisely what would change and what would not).

References

[1] D. Wu, L. Wang, P. Zhang, *Solving Statistical Mechanics Using Variational Autoregressive Networks*, Phys. Rev. Lett. **122**, 080602 (2019); arXiv:1809.10606.

[2] J. C. Wagner, A. Haghighat, *Automated Variance Reduction of Monte Carlo Shielding Calculations Using the Discrete Ordinates Adjoint Function (CADIS)*, Nucl. Sci. Eng. **128**, 186 (1998).

[3] J. C. Wagner, D. E. Peplow, S. W. Mosher, *FW-CADIS Method for Global and Regional Variance Reduction of Monte Carlo Radiation Transport Calculations*, Nucl. Sci. Eng. **176**, 37 (2014).

[4] M. A. Cooper, E. W. Larsen, *Automated Weight Windows for Global Monte Carlo Particle Transport*, Nucl. Sci. Eng. **137**, 1 (2001).

[5] D. M. Zuckerman, L. T. Chong, *Weighted Ensemble Simulation: Review of Methodology, Applications, and Software*, Annu. Rev. Biophys. **46**, 43 (2017).

[6] W. E, E. Vanden-Eijnden, *Towards a Theory of Transition Paths*, J. Stat. Phys. **123**, 503 (2006).

[7] *Nonequilibrium Statistical Mechanics Revealed by Doob Transform and Variational Autoregressive Networks*, Phys. Rev. E **111**, 034120 (2025).

[8] J. Spanier, E. M. Gelbard, *Monte Carlo Principles and Neutron Transport Problems* (1969).

[9] M. Kim, W. Cai, *Accelerated Markov Chain Monte Carlo simulation via neural network-driven importance sampling*, arXiv:2602.12294 (2026).

[10] L. Onsager, *Crystal statistics. I. A two-dimensional model with an order-disorder transition*, Phys. Rev. **65**, 117 (1944).

[11] A. L. Talapov, H. W. J. Blöte, *The magnetization of the 3D Ising model*, J. Phys. A **29**, 5727 (1996).

[12] K. Binder, A. P. Young, *Spin glasses: Experimental facts, theoretical concepts, and open questions*, Rev. Mod. Phys. **58**, 801 (1986).

[13] M. Hasenbusch, A. Pelissetto, E. Vicari, *Critical behavior of three-dimensional Ising spin glass models*, Phys. Rev. B **78**, 214205 (2008).

[14] L. Ts. Adzhemyan, D. A. Evdokimov, M. Hnatič, E. V. Ivanova, M. V. Kompaniets, A. Kudlis, D. V. Zakharov, *The dynamic critical exponent z for 2d and 3d Ising models from five-loop ε expansion*, Phys. Lett. A **425**, 127870 (2022).